

Class Notes III(Simplex Method)

Initial B.F.S., Simplex Table, and Computational Steps

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1 Basic idea and notation for Simplex Method

After introducing the necessary additional variables, write a linear programming problem in standard form as

$$\text{maximize } Z = c^T x, \quad Ax = b, \quad x \geq 0.$$

Choose m linearly independent columns of A to form the basis matrix B . The corresponding variables are basic; all remaining variables are non-basic. Putting the non-basic variables equal to zero gives $x_B = B^{-1}b$. This is a **basic feasible solution (B.F.S.)** when $x_B \geq 0$. If a basic variable is zero, the B.F.S. is degenerate.

Before beginning, make every b_i non-negative. If $b_i < 0$, multiply the whole constraint by -1 and reverse the inequality sign. For example,

$$7x_1 - 8x_2 \geq -3 \iff -7x_1 + 8x_2 \leq 3.$$

If the given problem is a minimization problem, maximize $Z' = -Z$. If $Z'_{\max} = v$, then $Z_{\min} = -v$.

2 Converting constraints and finding the initial B.F.S.

Table 1: Variables required for different constraints

Sign	Variable operation	Equation	Usual initial basic variable
$\leq$	Add slack variable $s_i \geq 0$	$a_i^T x + s_i = b_i$	s_i
$\geq$	Subtract surplus variable $s_i \geq 0$ and add artificial variable $A_i \geq 0$ when required	$a_i^T x - s_i + A_i = b_i$	A_i
$=$	Add artificial variable $A_i \geq 0$ if no suitable unit column already exists	$a_i^T x + A_i = b_i$	Existing unit-column variable or A_i

Meanings. A slack variable measures unused capacity. A surplus variable measures the amount above a minimum requirement. An artificial variable has no physical meaning; it is introduced only to construct an initial basis and must later be removed.

Table 2: Initial B.F.S. in the three main cases

Type of problem	Matrix and initial basis	Initial B.F.S.
All constraints $\leq$	$[A \mid I_m]$, with slack columns as $B = I_m$	Original variables = 0 and $s_i = b_i$
All constraints $\geq$	$[A \mid -I_m \mid I_m]$, with artificial columns as $B = I_m$	Original and surplus variables = 0 and $A_i = b_i$
Mixed $\leq, \geq$, and $=$	Use slack columns for $\leq$ rows and artificial or existing unit columns for the other rows	Non-basic variables = 0; each basic slack or artificial variable equals the corresponding b_i

3 Artificial variables explained with examples

An **artificial variable** is a temporary non-negative variable added with coefficient +1 when a constraint does not supply a convenient basic unit column. Its purpose is only to create an initial basis. It has no meaning in the original problem, so its value must become zero before the original problem is solved.

Example 1: a $\geq$ constraint

Consider

$$x_1 + 2x_2 \geq 6.$$

Subtracting a surplus variable gives

$$x_1 + 2x_2 - s_1 = 6.$$

If $x_1 = x_2 = 0$ and s_1 is chosen as the basic variable, then

$$-s_1 = 6 \implies s_1 = -6,$$

which violates $s_1 \geq 0$. Thus the surplus variable cannot give an initial B.F.S. Add an artificial variable A_1 :

$$\boxed{x_1 + 2x_2 - s_1 + A_1 = 6.}$$

Now the artificial column has coefficient +1, and the temporary initial B.F.S. is

$$x_1 = x_2 = s_1 = 0, \quad A_1 = 6.$$

This point is feasible only for the enlarged artificial problem. The simplex procedure must later make $A_1 = 0$.

Example 2: an equality constraint

Consider the two constraints

$$x_1 + x_2 \leq 4, \quad 2x_1 + x_2 = 5.$$

After adding a slack variable to the first row, its column is $[1, 0]^T$, but there is no second basic unit column. Add A_2 to the equality:

$$\begin{aligned} x_1 + x_2 + s_1 &= 4, \\ 2x_1 + x_2 + A_2 &= 5. \end{aligned}$$

The columns of s_1 and A_2 form I_2 . Therefore the temporary initial basis is $\{s_1, A_2\}$, and

$$x_1 = x_2 = 0, \quad s_1 = 4, \quad A_2 = 5.$$

Table 3: Why artificial variables are introduced

Constraint type	Problem without an artificial variable	Artificial-variable form
$a_i^T x \geq b_i$	The surplus coefficient is -1 ; using it as basic gives the value $-b_i$.	$a_i^T x - s_i + A_i = b_i$, initially $A_i = b_i$.
$a_i^T x = b_i$	No slack column is produced; a required basic unit column may be missing.	$a_i^T x + A_i = b_i$, initially $A_i = b_i$.

How artificial variables are removed

Suppose

$$\begin{aligned} \text{maximize } Z &= 3x_1 + 2x_2, \\ x_1 + x_2 &\leq 4, \\ x_1 + 3x_2 &\geq 6, \\ 2x_1 + x_2 &= 5, \quad x_1, x_2 \geq 0. \end{aligned}$$

The equations are

$$\begin{aligned} x_1 + x_2 + s_1 &= 4, \\ x_1 + 3x_2 - s_2 + A_2 &= 6, \\ 2x_1 + x_2 + A_3 &= 5. \end{aligned}$$

The initial basis is $\{s_1, A_2, A_3\}$, with

$$s_1 = 4, \quad A_2 = 6, \quad A_3 = 5.$$

1. **Two-phase method:** In Phase I, minimize

$$W = A_2 + A_3.$$

Here the initial value is $W = 6 + 5 = 11$. If the minimum becomes $W^* = 0$, then $A_2 = A_3 = 0$; delete their columns and maximize the original Z in Phase II. If $W^* > 0$, the original constraints are infeasible.

2. **Big- M method:** For the same maximization problem, use

$$Z_M = 3x_1 + 2x_2 - MA_2 - MA_3, \quad M > 0 \text{ very large.}$$

The large negative penalties force A_2 and A_3 toward zero. If an artificial variable is positive at the final optimum, the original problem is infeasible.

An artificial variable may remain basic with value zero because of degeneracy; this does not imply infeasibility. If the original coefficient matrix already contains a suitable basic unit column, no artificial variable is necessary for that row.

4 Simplex-table formulas

For the current basis B , define

$$x_B = B^{-1}b, \quad Y_j = B^{-1}\alpha_j, \quad Z = c_B^T x_B, \quad \boxed{\Delta_j = c_j - c_B^T Y_j}.$$

Here α_j is column j of the constraint matrix, Y_j is its transformed column, and c_B contains the objective coefficients of the basic variables.

Table 4: General simplex table

B	c_B	c_j x_B	c_1 $Y_1(\alpha_1)$	c_2 $Y_2(\alpha_2)$	$\cdots$ $\cdots$	c_n $Y_n(\alpha_n)$	c_{n+1} $Y_{n+1}(\beta_1)$	c_{n+2} $Y_{n+2}(\beta_2)$	$\cdots$ $\cdots$	c_N $Y_N(\beta_m)$	Ratio
$Y_{n+1}(\beta_1)$	c_{B1}	x_{B1}	y_{11}	y_{12}	$\cdots$	y_{1n}	1	0	$\cdots$	0	$\frac{x_{B1}}{y_{1k}}$
$Y_{n+2}(\beta_2)$	c_{B2}	x_{B2}	y_{21}	y_{22}	$\cdots$	y_{2n}	0	1	$\cdots$	0	$\frac{x_{B2}}{y_{2k}}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
$Y_{n+m}(\beta_m)$	c_{Bm}	x_{Bm}	y_{m1}	y_{m2}	$\cdots$	y_{mn}	0	0	$\cdots$	1	$\frac{x_{Bm}}{y_{mk}}$
$Z = c_B^T x_B$		Δ_j	Δ_1	Δ_2	$\cdots$	Δ_n	Δ_{n+1}	Δ_{n+2}	$\cdots$	Δ_{n+m}	Minimum

Here $y_{ij} = a_{ij}$ for all i and j . For a usual all- $\leq$ starting solution, the slack variables form the basis and $c_B = 0$. Therefore, initially,

$$Z = 0, \quad \Delta_j = c_j.$$

5 Optimality Test

The following table uses the convention $\Delta_j = c_j - c_B^T Y_j$ for a maximization problem.

Table 5: How to interpret the final row of the simplex table

Condition	Conclusion
$\Delta_j \leq 0$ for all j	The current B.F.S. is optimal.
Every non-basic $\Delta_j < 0$	The optimal solution is unique.
Some non-basic $\Delta_j = 0$ at an optimum	An alternative optimal solution may exist.
At least one $\Delta_j > 0$	The current solution is not optimal. In this case, we must proceed to improve the solution (step 5).
For an entering column with $\Delta_k > 0$, every $y_{ik} \leq 0$	No leaving variable exists; the maximization problem is unbounded.
Phase I ends with $W^* > 0$, or an artificial variable is positive at the Big- M optimum	The original problem has no feasible solution.

An artificial variable may remain basic at value zero because of degeneracy; this alone does not imply infeasibility.

6 Entering variable, leaving variable, and pivot

Table 6: Selection rules used in one simplex iteration

Item	Rule
Entering variable	Select α_k (<i>i.e.</i> x_k) corresponding to the largest positive reduced cost: $\Delta_k = \max\{\Delta_j : \Delta_j > 0\}$.
Leaving variable	Select row r using $\frac{x_{Br}}{y_{rk}} = \min_i \left\{ \frac{x_{Bi}}{y_{ik}} : y_{ik} > 0 \right\}$. Ignore zero and negative denominators.
Pivot element	The element y_{rk} at the intersection of the entering column and leaving row. Divide the pivot row by y_{rk} , then make all other entries in that column zero by row operations.

If the minimum ratio is tied, the next B.F.S. may be degenerate. If the largest positive Δ_j is tied, any tied column may be chosen; a consistent rule such as Bland's rule helps prevent cycling.

Entering variable, leaving variable, and pivot

Table 7: Selection rules used in one simplex iteration

Item	Rule
Entering variable	Select x_k corresponding to the largest positive reduced cost: $\Delta_k = \max\{\Delta_j : \Delta_j > 0\}$.
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If the minimum ratio is tied, the next B.F.S. may be degenerate. If the largest positive Δ_j is tied, any tied column may be chosen; a consistent rule such as Bland's rule helps prevent cycling.

7 Test the improved B.F.S. for optimality

If it is not optimal then repeat the steps 4, 5, 6 until we obtain an optimum solution.

Summary of the Above Steps of the Simplex Method for Maximization

Table 8: Complete computational procedure

Step	Procedure
1	If necessary, convert minimization into maximization by using $Z' = -Z$.
2	Make every right-hand side b_i non-negative. Multiply a negative row by -1 and reverse its inequality.
3	Convert every constraint into an equation using slack, surplus, and, when needed, artificial variables.
4	Find an initial basis and calculate $x_B = B^{-1}b$. If artificial variables occur, use the two-phase or Big- M method.
5	Construct the simplex table and calculate Y_j , Z , and every $\Delta_j = c_j - c_B^T Y_j$.
6	Test optimality. If all $\Delta_j \leq 0$, the current B.F.S. is optimal.
7	If some $\Delta_j > 0$, choose the column with the largest positive Δ_j as the entering column Y_k .
8	Choose the leaving row by the minimum positive-ratio test, pivot on the intersection entry, and construct the revised table.
9	Recalculate the reduced costs and repeat Steps 6–8 until an optimum is obtained or the problem is found to be unbounded or infeasible.

Example 1. Solve

$$\begin{aligned}
 &\text{Maximize} && Z = 40x_1 + 35x_2, \\
 &\text{subject to} && 2x_1 + 3x_2 \leq 60, \\
 & && 4x_1 + 3x_2 \leq 96, \\
 & && x_1, x_2 \geq 0.
 \end{aligned}$$

Solution: Step 1: Preparation.

The given problem is already a maximization problem and all the right-hand side constants are positive.

Step 2: Convert the constraints into equations.

Introduce slack variables $x_3, x_4 \geq 0$. Then

$$\begin{aligned}
 2x_1 + 3x_2 + x_3 &= 60, \\
 4x_1 + 3x_2 + x_4 &= 96.
 \end{aligned}$$

The objective function is $Z = 40x_1 + 35x_2 + 0x_3 + 0x_4$. Thus, $c_j = (40, 35, 0, 0)$.

Step 3: Initial basic feasible solution.

Initially, the slack variables x_3 and x_4 are basic. Setting the non-basic variables equal to zero, $x_1 = x_2 = 0$, we obtain $x_3 = 60$, $x_4 = 96$. Therefore, the initial B.F.S. is $(x_1, x_2, x_3, x_4) = (0, 0, 60, 96)$, and $Z = 0$. The initial basic vectors are $B = \{Y_3, Y_4\}$, with $c_B = (0, 0)$.

First Simplex Table

B	c_B	c_j	40	35	0	0	Min. ratio $x_B/Y_1, \quad y_{i1} > 0$
		x_B	Y_1	Y_2	$Y_3(\beta_1)$	$Y_4(\beta_2)$	
Y_3	0	60	2	3	1	0	$\frac{60}{2} = 30$
Y_4	0	96	4	3	0	1	$\frac{96}{4} = 24$ (min)
$Z = c_B^T x_B = 0$		Δ_j	40 $\uparrow$	35	0	0	x_1 enters

Step 4: Optimality test for the first table. Using $\Delta_j = c_j - c_B^T Y_j$, we have $\Delta_1 = 40 - (0, 0) \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 40$, and $\Delta_2 = 35 - (0, 0) \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 35$. Since $\Delta_1 = 40 > 0$, $\Delta_2 = 35 > 0$, the present solution is not optimal.

Step 5: Find the incoming and outgoing vectors.

Since $\Delta_1 = 40$ is the largest positive value, Y_1 is the incoming vector. For the Y_1 -column, $\frac{x_{B1}}{y_{11}} = \frac{60}{2} = 30$, and $\frac{x_{B2}}{y_{21}} = \frac{96}{4} = 24$. Hence, $\min\{30, 24\} = 24$. Therefore, Y_4 is the outgoing vector. Thus, the **pivot element**/ is $y_{21} = 4$.

Step 6: Row operations for the first pivot.

The old rows are $R_1 = [60, 2, 3, 1, 0]$, and $R_2 = [96, 4, 3, 0, 1]$. Since the pivot element is 4, divide the second row by 4: $R'_2 = \frac{1}{4}R_2 = \frac{1}{4}[96, 4, 3, 0, 1] = [24, 1, \frac{3}{4}, 0, \frac{1}{4}]$. Thus, $R'_2 = [24, 1, \frac{3}{4}, 0, \frac{1}{4}]$. Now eliminate the entry 2 from the Y_1 -column of the first row: $R'_1 = R_1 - 2R'_2 = [60, 2, 3, 1, 0] - 2[24, 1, \frac{3}{4}, 0, \frac{1}{4}] = [60 - 48, 2 - 2, 3 - \frac{3}{2}, 1, -\frac{1}{2}] = [12, 0, \frac{3}{2}, 1, -\frac{1}{2}]$. Thus, $R'_1 = [12, 0, \frac{3}{2}, 1, -\frac{1}{2}]$. The new basis is therefore $B = \{Y_3, Y_1\}$. The new cost vector of the basic variables is $c_B = (0, 40)$. The new objective value is $Z = (0)(12) + (40)(24) = 960$.

Second Simplex Table

B	c_B	c_j	40	35	0	0	Min. ratio $x_B/Y_2, \quad y_{i2} > 0$
		x_B	Y_1	Y_2	$Y_3(\beta_1)$	Y_4	
Y_3	0	12	0	$\frac{3}{2}$	1	$-\frac{1}{2}$	$\frac{12}{3/2} = 8$ (min)
Y_1	40	24	1	$\frac{3}{4}$	0	$\frac{1}{4}$	$\frac{24}{3/4} = 32$
$Z = c_B^T x_B = 960$		Δ_j	0	5 $\uparrow$	0	-10	x_2 enters

Step 7: Optimality test for the second table.

Since Y_1 and Y_3 are basic columns,

$$\Delta_1 = 0, \quad \Delta_3 = 0. \text{ For } Y_2, \Delta_2 = c_2 - c_B^T Y_2 = 35 - (0, 40) \begin{pmatrix} \frac{3}{2} \\ \frac{3}{4} \end{pmatrix} = 35 - 30 = 5.$$

For Y_4 , $\Delta_4 = c_4 - c_B^T Y_4 = 0 - (0, 40) \begin{pmatrix} -\frac{1}{2} \\ \frac{1}{4} \end{pmatrix} = -10$. Therefore, $(\Delta_1, \Delta_2, \Delta_3, \Delta_4) = (0, 5, 0, -10)$.

Since $\Delta_2 = 5 > 0$, the solution is not yet optimal.

Step 8: Find the incoming and outgoing vectors.

Since Δ_2 is the maximum value, Y_2 is the incoming vector.

For the minimum ratio test, $\frac{12}{3/2} = 8$, and $\frac{24}{3/4} = 32$. Thus, $\min\{8, 32\} = 8$. Therefore, Y_3 is the outgoing vector.

The key element is $y_{12} = \frac{3}{2}$.

Step 9: Row operations for the second pivot.

First divide the first row by $\frac{3}{2}$:

$$R_1'' = \frac{R_1'}{3/2} = \frac{2}{3} R_1' = \frac{2}{3} \left[12, 0, \frac{3}{2}, 1, -\frac{1}{2} \right] = \left[8, 0, 1, \frac{2}{3}, -\frac{1}{3} \right].$$

Thus, $R_1'' = [8, 0, 1, \frac{2}{3}, -\frac{1}{3}]$. Now eliminate $\frac{3}{4}$ from the Y_2 -column of the second row:

$$R_2'' = R_2' - \frac{3}{4} R_1'' = \left[24, 1, \frac{3}{4}, 0, \frac{1}{4} \right] - \frac{3}{4} \left[8, 0, 1, \frac{2}{3}, -\frac{1}{3} \right] = \left[24 - 6, 1, 0, -\frac{1}{2}, \frac{1}{2} \right] = \left[18, 1, 0, -\frac{1}{2}, \frac{1}{2} \right].$$

Therefore, $R_2'' = [18, 1, 0, -\frac{1}{2}, \frac{1}{2}]$. For the reduced-cost row, $R_\Delta'' = R_\Delta' - 5R_1'' = [0, 5, 0, -10] - 5[0, 1, \frac{2}{3}, -\frac{1}{3}] = [0, 0, -\frac{10}{3}, -\frac{25}{3}]$. The new objective value can be computed as

$$Z = 35(8) + 40(18) = 280 + 720 = 1000.$$

Third Simplex Table

B	c_B	c_j	40	35	0	0	Status
		x_B	Y_1	Y_2	Y_3	Y_4	
Y_2	35	8	0	1	$\frac{2}{3}$	$-\frac{1}{3}$	—
Y_1	40	18	1	0	$-\frac{1}{2}$	$\frac{1}{2}$	—
$Z = c_B^T x_B = 1000$		Δ_j	0	0	$-\frac{10}{3}$	$-\frac{25}{3}$	Optimal

Step 10: Optimality test.

In the final table, $\Delta_1 = 0$, $\Delta_2 = 0$, $\Delta_3 = -\frac{10}{3}$, $\Delta_4 = -\frac{25}{3}$. Hence, $\Delta_j \leq 0$ for all j . Therefore, the current solution is optimal. The basic variables are x_2 and x_1 . Hence, $x_2 = 8$, $x_1 = 18$. The non-basic variables are $x_3 = x_4 = 0$.

Therefore, $x_1 = 18$, $x_2 = 8$. The maximum value of the objective function is $Z_{\max} = 40(18) + 35(8) = 720 + 280 = 1000$. Thus, $x_1 = 18$, $x_2 = 8$, $Z_{\max} = 1000$.

Moreover, $\Delta_3 = -\frac{10}{3} < 0$, $\Delta_4 = -\frac{25}{3} < 0$, for both non-basic variables. Hence, the obtained solution is the **unique optimal solution**.

The above solution with its different computational steps can be more conveniently represented by a single table as shown below:

Table 9: Combined simplex table showing all iterations

B	c_B	c_j x_B	40 Y_1	35 Y_2	0 Y_3	0 Y_4	Minimum ratio Entering column
Y_3	0	60	2	3	1	0	$60/2 = 30$
Y_4	0	96	4	3	0	1	$96/4 = 24$ (minimum)
$Z = 0$		Δ_j	40	35	0	0	x_1 enters
Y_2	0	12	0	$3/2$	1	$-1/2$	$12/(3/2) = 8$ (minimum)
Y_1	40	24	1	$3/4$	0	$1/4$	$24/(3/4) = 32$
$Z = 960$		Δ_j	0	5	0	-10	x_2 enters
Y_2	35	8	0	1	$2/3$	$-1/3$	—
Y_1	40	18	1	0	$-1/2$	$1/2$	—
$Z = 1000$		Δ_j	0	0	$-10/3$	$-25/3$	Optimal

First pivot. In the first block, the largest positive reduced cost is $\Delta_1 = 40$; therefore x_1 enters. The ratios are 30 and 24, so x_4 leaves and the pivot element is 4. Perform

$$R_2 \leftarrow \frac{1}{4}R_2, \quad R_1 \leftarrow R_1 - 2R_2.$$

This produces the second block with basis $\{x_3, x_1\}$. The objective coefficient over the x_1 column remains 40, and

$$Z = (0)(12) + (40)(24) = 960,$$

$$\Delta_2 = 35 - (0, 40) \begin{bmatrix} 3/2 \\ 3/4 \end{bmatrix} = 5,$$

$$\Delta_4 = 0 - (0, 40) \begin{bmatrix} -1/2 \\ 1/4 \end{bmatrix} = -10.$$

Second pivot. Since $\Delta_2 = 5 > 0$, x_2 enters. The ratios are 8 and 32, so x_3 leaves and the pivot element is $3/2$. Perform

$$R_1 \leftarrow \frac{2}{3}R_1, \quad R_2 \leftarrow R_2 - \frac{3}{4}R_1.$$

The third block has basis $\{x_2, x_1\}$ and represents the final tableau.

Now every $\Delta_j \leq 0$, so the current B.F.S. is optimal. Moreover, $\Delta_3 = -10/3 < 0$ and $\Delta_4 = -25/3 < 0$ for the non-basic variables x_3, x_4 ; hence the optimum is unique. Reading the basic variables from the last block of the combined table and setting the non-basic variables to zero gives

$$\boxed{x_1 = 18, \quad x_2 = 8, \quad Z_{\max} = 1000.}$$

Example 2. Solve by the simplex method:

$$\begin{aligned} \text{Maximize } & Z = 3x_1 + 5x_2 + 4x_3, \\ \text{subject to } & 2x_1 + 3x_2 \leq 8, \\ & 2x_2 + 5x_3 \leq 10, \\ & 3x_1 + 2x_2 + 4x_3 \leq 15, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Solution: Step 1: Preparation.

The given problem is already a maximization problem and all the right-hand side constants are positive.

Step 2: Convert the constraints into standard form.

Introduce the slack variables

$$x_4, x_5, x_6 \geq 0.$$

Then the constraints become

$$\begin{aligned} 2x_1 + 3x_2 + x_4 &= 8, \\ 2x_2 + 5x_3 + x_5 &= 10, \\ 3x_1 + 2x_2 + 4x_3 + x_6 &= 15. \end{aligned}$$

$$Z = 3x_1 + 5x_2 + 4x_3 + 0x_4 + 0x_5 + 0x_6. \text{ Therefore, } c_j = (3, 5, 4, 0, 0, 0).$$

Step 3: Initial basic feasible solution.

Initially, the slack variables x_4, x_5, x_6 are basic variables.

Putting the non-basic variables equal to zero, $x_1 = x_2 = x_3 = 0$, we obtain $x_4 = 8$, $x_5 = 10$, $x_6 = 15$. Hence, the initial B.F.S. is $(x_1, x_2, x_3, x_4, x_5, x_6) = (0, 0, 0, 8, 10, 15)$, with $Z = 0$. The initial basis is $B = \{Y_4, Y_5, Y_6\}$, and $c_B = (0, 0, 0)$.

First Simplex Table

B	c_B	c_j	3	5	4	0	0	0	Min. ratio $x_B/Y_2, \quad y_{i2} > 0$
		x_B	Y_1	Y_2	Y_3	$Y_4(\beta_1)$	$Y_5(\beta_2)$	$Y_6(\beta_3)$	
Y_4	0	8	2	3	0	1	0	0	$\frac{8}{3}$ (min) $\frac{10}{2} = 5$ $\frac{15}{2}$
Y_5	0	10	0	2	5	0	1	0	
Y_6	0	15	3	2	4	0	0	1	
$Z = c_B^T x_B = 0$		Δ_j	3	5 $\uparrow$	4	0	0	0	x_2 enters

Step 4: Optimality test for the first simplex table.

Using $\Delta_j = c_j - c_B^T Y_j$, we obtain

$$\Delta_1 = 3 - (0, 0, 0) \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix} = 3, \Delta_2 = 5 - (0, 0, 0) \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} = 5, \text{ and } \Delta_3 = 4 - (0, 0, 0) \begin{pmatrix} 0 \\ 5 \\ 4 \end{pmatrix} = 4.$$

Thus, $\Delta_1 = 3$, $\Delta_2 = 5$, $\Delta_3 = 4$. Since some $\Delta_j > 0$, the present solution is not optimal.

Step 5: Find the incoming and outgoing vectors.

Since $\Delta_2 = 5$ is the maximum positive value of Δ_j , Y_2 is the incoming vector.

For the Y_2 -column, the ratios are $\frac{x_{B1}}{y_{12}} = \frac{8}{3}$, $\frac{x_{B2}}{y_{22}} = \frac{10}{2} = 5$, and $\frac{x_{B3}}{y_{32}} = \frac{15}{2}$.

Therefore, $\min \left\{ \frac{8}{3}, 5, \frac{15}{2} \right\} = \frac{8}{3}$. Hence, Y_4 is the outgoing vector. The pivot element is $y_{12} = 3$.

Step 6: Row operations for the first pivot.

The original rows are

$$R_1 = [8, 2, 3, 0, 1, 0, 0], R_2 = [10, 0, 2, 5, 0, 1, 0], \text{ and } R_3 = [15, 3, 2, 4, 0, 0, 1].$$

Since the pivot element is 3, divide the first row by 3:

$$R'_1 = \frac{1}{3}R_1 = \frac{1}{3}[8, 2, 3, 0, 1, 0, 0] = \left[\frac{8}{3}, \frac{2}{3}, 1, 0, \frac{1}{3}, 0, 0 \right].$$

Hence, $R'_1 = \left[\frac{8}{3}, \frac{2}{3}, 1, 0, \frac{1}{3}, 0, 0 \right]$. Now eliminate the entry 2 from the Y_2 -column of the second row:

$$R'_2 = R_2 - 2R'_1 = [10, 0, 2, 5, 0, 1, 0] - 2 \left[\frac{8}{3}, \frac{2}{3}, 1, 0, \frac{1}{3}, 0, 0 \right] = \left[\frac{14}{3}, -\frac{4}{3}, 0, 5, -\frac{2}{3}, 1, 0 \right].$$

Thus, $R'_2 = \left[\frac{14}{3}, -\frac{4}{3}, 0, 5, -\frac{2}{3}, 1, 0 \right]$. Similarly,

$$R'_3 = R_3 - 2R'_1 = [15, 3, 2, 4, 0, 0, 1] - 2 \left[\frac{8}{3}, \frac{2}{3}, 1, 0, \frac{1}{3}, 0, 0 \right] = \left[\frac{29}{3}, \frac{5}{3}, 0, 4, -\frac{2}{3}, 0, 1 \right].$$

Therefore, $R'_3 = \left[\frac{29}{3}, \frac{5}{3}, 0, 4, -\frac{2}{3}, 0, 1 \right]$. The new basis is $B = \{Y_2, Y_5, Y_6\}$, with $c_B = (5, 0, 0)$.

The new objective value is $Z = 5 \left(\frac{8}{3} \right) = \frac{40}{3}$.

Second Simplex Table

B	c_B	c_j	3	5	4	0	0	0	Min. ratio $x_B/Y_3, \quad y_{i3} > 0$
		x_B	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	
Y_2	5	$\frac{8}{3}$	$\frac{2}{3}$	1	0	$\frac{1}{3}$	0	0	—
Y_5	0	$\frac{14}{3}$	$-\frac{4}{3}$	0	$\frac{5}{3}$	$-\frac{2}{3}$	1	0	$\frac{14/3}{5} = \frac{14}{15}$ (min)
Y_6	0	$\frac{29}{3}$	$\frac{5}{3}$	0	4	$-\frac{2}{3}$	0	1	$\frac{29/3}{4} = \frac{29}{12}$
$Z = c_B^T x_B = \frac{40}{3}$		Δ_j	$-\frac{1}{3}$	0	4 $\uparrow$	$-\frac{5}{3}$	0	0	x_3 enters

Step 7: Optimality test for the second simplex table.

Since Y_2, Y_5, Y_6 are basic columns, $\Delta_2 = \Delta_5 = \Delta_6 = 0$.

$$\text{For } Y_1, \quad \Delta_1 = 3 - (5, 0, 0) \begin{pmatrix} \frac{2}{3} \\ -\frac{4}{3} \\ \frac{5}{3} \end{pmatrix} = 3 - \frac{10}{3} = -\frac{1}{3}. \quad \text{For } Y_3, \quad \Delta_3 = 4 - (5, 0, 0) \begin{pmatrix} 0 \\ 5 \\ 4 \end{pmatrix} = 4. \quad \text{For } Y_4,$$

$$\Delta_4 = 0 - (5, 0, 0) \begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \\ -\frac{2}{3} \end{pmatrix} = -\frac{5}{3}. \quad \text{Therefore, } (\Delta_1, \Delta_2, \Delta_3, \Delta_4, \Delta_5, \Delta_6) = \left(-\frac{1}{3}, 0, 4, -\frac{5}{3}, 0, 0\right).$$

Since $\Delta_3 = 4 > 0$, the solution is not optimal.

Step 8: Find the incoming and outgoing vectors.

Since $\Delta_3 = 4$ is the maximum positive value, Y_3 is the incoming vector.

For the Y_3 -column, the first row is zero and is not considered. For the remaining positive entries, $\frac{x_{B_2}}{y_{23}} = \frac{14/3}{5} = \frac{14}{15}$, and $\frac{x_{B_3}}{y_{33}} = \frac{29/3}{4} = \frac{29}{12}$. Hence, $\min \left\{ \frac{14}{15}, \frac{29}{12} \right\} = \frac{14}{15}$. Therefore, Y_5 is the outgoing vector. The pivot element is $y_{23} = 5$.

Step 9: Row operations for the second pivot.

Divide the second row by 5:

$$R_2'' = \frac{1}{5} R_2' = \frac{1}{5} \left[\frac{14}{3}, -\frac{4}{3}, 0, 5, -\frac{2}{3}, 1, 0 \right] = \left[\frac{14}{15}, -\frac{4}{15}, 0, 1, -\frac{2}{15}, \frac{1}{5}, 0 \right].$$

Thus, $R_2'' = \left[\frac{14}{15}, -\frac{4}{15}, 0, 1, -\frac{2}{15}, \frac{1}{5}, 0 \right]$. Since the coefficient of Y_3 in the first row is zero,

$$R_1'' = R_1' = \left[\frac{8}{3}, \frac{2}{3}, 1, 0, \frac{1}{3}, 0, 0 \right].$$

Now eliminate 4 from the Y_3 -column of the third row:

$$R_3'' = R_3' - 4R_2'' = \left[\frac{29}{3}, \frac{5}{3}, 0, 4, -\frac{2}{3}, 0, 1 \right] - 4 \left[\frac{14}{15}, -\frac{4}{15}, 0, 1, -\frac{2}{15}, \frac{1}{5}, 0 \right] = \left[\frac{89}{15}, \frac{41}{15}, 0, 0, -\frac{2}{15}, -\frac{4}{5}, 1 \right].$$

Therefore, $R_3'' = \left[\frac{89}{15}, \frac{41}{15}, 0, 0, -\frac{2}{15}, -\frac{4}{5}, 1 \right]$. The new basis is $B = \{Y_2, Y_3, Y_6\}$, with $c_B = (5, 4, 0)$. The objective value is

$$Z = 5 \left(\frac{8}{3} \right) + 4 \left(\frac{14}{15} \right) = \frac{40}{3} + \frac{56}{15} = \frac{256}{15}.$$

Third Simplex Table

B	c_B	c_j	3	5	4	0	0	0	Min. ratio $x_B/Y_1, \quad y_{i1} > 0$
		x_B	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	
Y_2	5	$\frac{8}{3}$	$\frac{2}{3}$	1	0	$\frac{1}{3}$	0	0	$\frac{8/3}{2/3} = 4$
Y_3	4	$\frac{14}{15}$	$-\frac{4}{15}$	0	1	$-\frac{2}{15}$	$\frac{1}{5}$	0	Neg.
Y_6	0	$\frac{89}{15}$	$\frac{41}{15}$	0	0	$-\frac{2}{15}$	$-\frac{4}{5}$	1	$\frac{89/15}{41/15} = \frac{89}{41}$ (min)
$Z = c_B^T x_B = \frac{256}{15}$		Δ_j	$\frac{11}{15}$ ↑	0	0	$-\frac{17}{15}$	$-\frac{4}{5}$	0	x_1 enters

Step 10: Optimality test for the third simplex table.

Since Y_2, Y_3, Y_6 are basic columns, $\Delta_2 = \Delta_3 = \Delta_6 = 0$.

$$\text{For } Y_1, \Delta_1 = 3 - (5, 4, 0) \begin{pmatrix} \frac{2}{3} \\ \frac{4}{15} \\ \frac{41}{15} \end{pmatrix} = 3 - \left(\frac{10}{3} - \frac{16}{15}\right) = 3 - \frac{34}{15} = \frac{11}{15}.$$

$$\text{For } Y_4, \Delta_4 = 0 - (5, 4, 0) \begin{pmatrix} \frac{1}{3} \\ \frac{2}{15} \\ \frac{2}{15} \end{pmatrix} = -\frac{17}{15}. \text{ For } Y_5, \Delta_5 = 0 - (5, 4, 0) \begin{pmatrix} 0 \\ \frac{1}{5} \\ -\frac{4}{5} \end{pmatrix} = -\frac{4}{5}.$$

Thus, $(\Delta_1, \Delta_2, \Delta_3, \Delta_4, \Delta_5, \Delta_6) = (\frac{11}{15}, 0, 0, -\frac{17}{15}, -\frac{4}{5}, 0)$. Since $\Delta_1 = \frac{11}{15} > 0$, the solution is not yet optimal.

Step 11: Find the incoming and outgoing vectors.

Since $\Delta_1 = \frac{11}{15}$ is the maximum positive value, $\boxed{Y_1 \text{ is the incoming vector.}}$ For the Y_1 -column, the positive entries give $\frac{8/3}{2/3} = 4$, and $\frac{89/15}{41/15} = \frac{89}{41}$. The second row is not considered because $-\frac{4}{15} < 0$.

Thus, $\min\{4, \frac{89}{41}\} = \frac{89}{41}$. Therefore, $\boxed{Y_6 \text{ is the outgoing vector.}}$ The pivot element is $y_{31} = \frac{41}{15}$.

Step 12: Row operations for the third pivot.

First divide the third row by $\frac{41}{15}$:

$$R_3''' = \frac{15}{41}R_3'' = \frac{15}{41} \left[\frac{89}{15}, \frac{41}{15}, 0, 0, -\frac{2}{15}, -\frac{4}{5}, 1 \right] = \left[\frac{89}{41}, 1, 0, 0, -\frac{2}{41}, -\frac{12}{41}, \frac{15}{41} \right].$$

Thus, $R_3''' = [\frac{89}{41}, 1, 0, 0, -\frac{2}{41}, -\frac{12}{41}, \frac{15}{41}]$. Now eliminate $\frac{2}{3}$ from the Y_1 -column of the first row:

$$R_1''' = R_1'' - \frac{2}{3}R_3''' = \left[\frac{8}{3}, \frac{2}{3}, 1, 0, \frac{1}{3}, 0, 0 \right] - \frac{2}{3} \left[\frac{89}{41}, 1, 0, 0, -\frac{2}{41}, -\frac{12}{41}, \frac{15}{41} \right] = \left[\frac{50}{41}, 0, 1, 0, \frac{15}{41}, \frac{8}{41}, -\frac{10}{41} \right].$$

Therefore, $R_1''' = [\frac{50}{41}, 0, 1, 0, \frac{15}{41}, \frac{8}{41}, -\frac{10}{41}]$. The coefficient of Y_1 in the second row is $-\frac{4}{15}$. Hence,

$$R_2''' = R_2'' + \frac{4}{15}R_3''' = \left[\frac{14}{15}, -\frac{4}{15}, 0, 1, -\frac{2}{15}, \frac{1}{5}, 0 \right] + \frac{4}{15} \left[\frac{89}{41}, 1, 0, 0, -\frac{2}{41}, -\frac{12}{41}, \frac{15}{41} \right] = \left[\frac{62}{41}, 0, 0, 1, -\frac{6}{41}, \frac{5}{41}, \frac{4}{41} \right].$$

Hence, $R_2''' = [\frac{62}{41}, 0, 0, 1, -\frac{6}{41}, \frac{5}{41}, \frac{4}{41}]$. The new basis is $\boxed{B = \{Y_2, Y_3, Y_1\}}$, with $c_B = (5, 4, 3)$.

The new objective value is

$$Z = 5 \left(\frac{50}{41} \right) + 4 \left(\frac{62}{41} \right) + 3 \left(\frac{89}{41} \right) = \frac{250 + 248 + 267}{41} = \frac{765}{41}.$$

Fourth Simplex Table

B	c_B	c_j	3	5	4	0	0	0	Status
		x_B	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	
Y_2	5	$\frac{50}{41}$	0	1	0	$\frac{15}{41}$	$\frac{8}{41}$	$-\frac{10}{41}$	—
Y_3	4	$\frac{62}{41}$	0	0	1	$-\frac{6}{41}$	$\frac{5}{41}$	$\frac{4}{41}$	—
Y_1	3	$\frac{89}{41}$	1	0	0	$-\frac{2}{41}$	$-\frac{12}{41}$	$\frac{15}{41}$	—
$Z = c_B^T x_B = \frac{765}{41}$		Δ_j	0	0	0	$-\frac{45}{41}$	$-\frac{24}{41}$	$-\frac{11}{41}$	Optimal

Step 13: Final optimality test.

From the fourth simplex table, $\Delta_1 = 0$, $\Delta_2 = 0$, $\Delta_3 = 0$, and $\Delta_4 = -\frac{45}{41}$, $\Delta_5 = -\frac{24}{41}$, $\Delta_6 = -\frac{11}{41}$. Hence, $\Delta_j \leq 0$ for all j . Therefore, the present solution is optimal.

The basic variables in the final table are Y_2 , Y_3 , Y_1 . Hence, $x_2 = \frac{50}{41}$, $x_3 = \frac{62}{41}$, $x_1 = \frac{89}{41}$. The non-basic variables are $x_4 = x_5 = x_6 = 0$. Therefore,

$$x_1 = \frac{89}{41}, \quad x_2 = \frac{50}{41}, \quad x_3 = \frac{62}{41}.$$

The maximum value of the objective function is

$$Z_{\max} = 3 \left(\frac{89}{41} \right) + 5 \left(\frac{50}{41} \right) + 4 \left(\frac{62}{41} \right) = \frac{267 + 250 + 248}{41} = \frac{765}{41}.$$

Thus, the required optimal solution is

$$\boxed{x_1 = \frac{89}{41}, \quad x_2 = \frac{50}{41}, \quad x_3 = \frac{62}{41}, \quad Z_{\max} = \frac{765}{41}}.$$

Moreover, for all the non-basic variables,

$$\Delta_4 = -\frac{45}{41} < 0, \quad \Delta_5 = -\frac{24}{41} < 0, \quad \Delta_6 = -\frac{11}{41} < 0.$$

Hence, the obtained solution is the **unique optimal solution**.

All four simplex tableaux are represented by the following *single combined table*. Each thick horizontal division begins a new iteration.

Table 10: Single combined simplex table for all four iterations

B	c_B	c_j X_B	3 Y_1	5 Y_2	4 Y_3	0 $Y_4(\beta_1)$	0 $Y_5(\beta_2)$	0 $Y_6(\beta_3)$	Minimum ratio Entering column
Y_4	0	8	2	3	0	1	0	0	8/3 (minimum)
Y_5	0	10	0	2	5	0	1	0	10/2 = 5
Y_6	0	15	3	2	4	0	0	1	15/2
$Z = 0$		Δ_j	3	5	4	0	0	0	x_2 enters
Y_2	5	8/3	2/3	1	0	1/3	0	0	—
Y_5	0	14/3	-4/3	0	5	-2/3	1	0	(14/3)/5 = 14/15 (minimum)
Y_6	0	29/3	5/3	0	4	-2/3	0	1	(29/3)/4 = 29/12
$Z = 40/3$		Δ_j	-1/3	0	4	-5/3	0	0	x_3 enters
Y_2	5	8/3	2/3	1	0	1/3	0	0	(8/3)/(2/3) = 4
Y_3	4	14/15	-4/15	0	1	-2/15	1/5	0	—
Y_6	0	89/15	41/15	0	0	-2/15	-4/5	1	(89/15)/(41/15) = 89/41 (minimum)
$Z = 256/15$		Δ_j	11/15	0	0	-17/15	-4/5	0	x_1 enters
Y_2	5	50/41	0	1	0	15/41	8/41	-10/41	—
Y_3	4	62/41	0	0	1	-6/41	5/41	4/41	—
Y_1	3	89/41	1	0	0	-2/41	-12/41	15/41	—
$Z = 765/41$		Δ_j	0	0	0	-45/41	-24/41	-11/41	Optimal

The successive pivot decisions are summarized below:

1. $\Delta_2 = 5$ is largest, so x_2 enters. The minimum ratio is $8/3$, so x_4 leaves; pivot = 3. To obtain the second block, use

$$R_1 \leftarrow \frac{1}{3}R_1, \quad R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 - 2R_1.$$

2. In the second block, $\Delta_3 = 4$, so x_3 enters. The minimum ratio is $14/15$, so x_5 leaves; pivot = 5. To obtain the third block, use

$$R_2 \leftarrow \frac{1}{5}R_2, \quad R_3 \leftarrow R_3 - 4R_2.$$

The first row is unchanged because its x_3 coefficient is zero.

3. In the third block, $\Delta_1 = 11/15$, so x_1 enters. The minimum ratio is $89/41$, so x_6 leaves; pivot = $41/15$. To obtain the final block, use

$$R_3 \leftarrow \frac{15}{41}R_3, \quad R_1 \leftarrow R_1 - \frac{2}{3}R_3, \quad R_2 \leftarrow R_2 + \frac{4}{15}R_3.$$

In the last block all reduced costs satisfy $\Delta_j \leq 0$. Thus

$x_1 = \frac{89}{41},$	$x_2 = \frac{50}{41},$	$x_3 = \frac{62}{41},$	$Z_{\max} = \frac{765}{41}.$
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Since the reduced costs of all non-basic variables x_4, x_5, x_6 are strictly negative, this optimal solution is unique.

Quick revision box

$$x_B = B^{-1}b, \quad Y_j = B^{-1}a_j, \\ Z = c_B^T x_B, \quad \Delta_j = c_j - c_B^T Y_j.$$

For maximization: largest positive Δ_j enters; the minimum positive ratio determines the leaving row; pivot and repeat. Stop when every $\Delta_j \leq 0$.

Example 3: Solve by simplex method the following L.P. problem:

$$\text{Minimize} \quad Z = x_1 - 3x_2 + 2x_3,$$

subject to

$$\begin{aligned} 3x_1 - x_2 + 2x_3 &\leq 7, \\ -2x_1 + 4x_2 &\leq 12, \\ -4x_1 + 3x_2 + 8x_3 &\leq 10, \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

Solution: Step 1: Convert the minimization problem into a maximization problem.

Since the simplex method is being applied in maximization form, put $Z' = -Z$. Therefore,

$$\text{Maximize} \quad Z' = -x_1 + 3x_2 - 2x_3.$$

Step 2: Convert the constraints into standard form.

Introduce the slack variables $x_4, x_5, x_6 \geq 0$. Then

$$\begin{aligned} 3x_1 - x_2 + 2x_3 + x_4 &= 7, \\ -2x_1 + 4x_2 + x_5 &= 12, \\ -4x_1 + 3x_2 + 8x_3 + x_6 &= 10. \end{aligned}$$

The objective function becomes $Z' = -x_1 + 3x_2 - 2x_3 + 0x_4 + 0x_5 + 0x_6$. Thus, $c_j = (-1, 3, -2, 0, 0, 0)$.

Step 3: Find the initial basic feasible solution.

Initially, the slack variables x_4, x_5, x_6 are basic variables.

Setting the non-basic variables equal to zero, $x_1 = x_2 = x_3 = 0$, we get $x_4 = 7$, $x_5 = 12$, $x_6 = 10$. Hence the initial B.F.S. is $(x_1, x_2, x_3, x_4, x_5, x_6) = (0, 0, 0, 7, 12, 10)$, and $Z' = 0$. The initial basis is $B = \{Y_4, Y_5, Y_6\}$, with $c_B = (0, 0, 0)$.

First Simplex Table

B	c_B	c_j	-1	3	-2	0	0	0	Min. ratio $x_B/Y_2, \quad y_{i2} > 0$
		x_B	Y_1	Y_2	Y_3	$Y_4(\beta_1)$	$Y_5(\beta_2)$	$Y_6(\beta_3)$	
Y_4	0	7	3	-1	2	1	0	0	Neg. $\frac{12}{4} = 3$ (min) $\frac{10}{3}$
Y_5	0	12	-2	4	0	0	1	0	
Y_6	0	10	-4	3	8	0	0	1	
$Z' = c_B^T x_B = 0$		Δ_j	-1	3 ↑	-2	0	0	0	x_2 enters

Step 4: Optimality test for the first simplex table.

Using $\Delta_j = c_j - c_B^T Y_j$, we obtain $\Delta_1 = -1 - (0, 0, 0) \begin{pmatrix} 3 \\ -2 \\ -4 \end{pmatrix} = -1$, $\Delta_2 = 3 - (0, 0, 0) \begin{pmatrix} -1 \\ 4 \\ 3 \end{pmatrix} = 3$,

and $\Delta_3 = -2 - (0, 0, 0) \begin{pmatrix} 2 \\ 0 \\ 8 \end{pmatrix} = -2$. Thus, $\Delta_1 = -1$, $\Delta_2 = 3$, $\Delta_3 = -2$. Since $\Delta_2 = 3 > 0$,

the present solution is not optimal.

Step 5: Find the incoming and outgoing vectors.

Since $\Delta_2 = 3$ is the maximum positive value of Δ_j , Y_2 is the incoming vector. For the Y_2 -column, only positive entries are considered.

The first entry is negative, $y_{12} = -1 < 0$, so the first row is not considered. For the second row, $\frac{x_{B_2}}{y_{22}} = \frac{12}{4} = 3$. For the third row, $\frac{x_{B_3}}{y_{32}} = \frac{10}{3}$. Hence, $\min \{3, \frac{10}{3}\} = 3$. Therefore, Y_5 is the outgoing vector. Thus, the pivot element is $y_{22} = 4$.

Step 6: Perform the first pivot row operations.

The original rows are $R_1 = [7, 3, -1, 2, 1, 0, 0]$, $R_2 = [12, -2, 4, 0, 0, 1, 0]$, and $R_3 = [10, -4, 3, 8, 0, 0, 1]$. Since the pivot element is 4, divide the second row by 4:

$$R'_2 = \frac{1}{4}R_2 = \frac{1}{4}[12, -2, 4, 0, 0, 1, 0] = \left[3, -\frac{1}{2}, 1, 0, 0, \frac{1}{4}, 0\right].$$

Thus, $R'_2 = [3, -\frac{1}{2}, 1, 0, 0, \frac{1}{4}, 0]$. Now eliminate the entry -1 from the Y_2 -column of the first row. Since the coefficient is -1 , $R'_1 = R_1 + R'_2$. Hence,
 $R'_1 = [7, 3, -1, 2, 1, 0, 0] + [3, -\frac{1}{2}, 1, 0, 0, \frac{1}{4}, 0] = [10, \frac{5}{2}, 0, 2, 1, \frac{1}{4}, 0]$. Therefore, $R'_1 = [10, \frac{5}{2}, 0, 2, 1, \frac{1}{4}, 0]$.
 Now eliminate 3 from the Y_2 -column of the third row:

$$R'_3 = R_3 - 3R'_2 = [10, -4, 3, 8, 0, 0, 1] - 3 \left[3, -\frac{1}{2}, 1, 0, 0, \frac{1}{4}, 0 \right] = \left[1, -\frac{5}{2}, 0, 8, 0, -\frac{3}{4}, 1 \right].$$

Hence, $R'_3 = [1, -\frac{5}{2}, 0, 8, 0, -\frac{3}{4}, 1]$. The new basis is $B = \{Y_4, Y_2, Y_6\}$, and $c_B = (0, 3, 0)$. The new objective value is $Z' = (0)(10) + (3)(3) + (0)(1) = 9$.

Second Simplex Table

B	c_B	c_j	-1	3	-2	0	0	0	Min. ratio $x_B/Y_1, \quad y_{i1} > 0$
		x_B	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	
Y_4	0	10	$\frac{5}{2}$	0	2	1	$\frac{1}{4}$	0	$\frac{10}{5/2} = 4$ (min)
Y_2	3	3	$-\frac{1}{2}$	1	0	0	$\frac{1}{4}$	0	Neg.
Y_6	0	1	$-\frac{5}{2}$	0	8	0	$-\frac{3}{4}$	1	Neg.
$Z' = c_B^T x_B = 9$		Δ_j	$\frac{1}{2}$ $\uparrow$	0	-2	0	$-\frac{3}{4}$	0	x_1 enters

Step 7: Optimality test for the second simplex table.

Since Y_2, Y_4, Y_6 are basic columns, $\Delta_2 = \Delta_4 = \Delta_6 = 0$.

For Y_1 , $\Delta_1 = -1 - (0, 3, 0) \begin{pmatrix} \frac{5}{2} \\ -\frac{1}{2} \\ \frac{5}{2} \\ -\frac{5}{2} \end{pmatrix} = -1 - (-\frac{3}{2}) = \frac{1}{2}$. For Y_3 , $\Delta_3 = -2 - (0, 3, 0) \begin{pmatrix} 2 \\ 0 \\ 8 \end{pmatrix} = -2$.

For Y_5 , $\Delta_5 = 0 - (0, 3, 0) \begin{pmatrix} \frac{1}{4} \\ \frac{1}{4} \\ \frac{3}{4} \\ -\frac{3}{4} \end{pmatrix} = -\frac{3}{4}$. Therefore, $(\Delta_1, \Delta_2, \Delta_3, \Delta_4, \Delta_5, \Delta_6) = (\frac{1}{2}, 0, -2, 0, -\frac{3}{4}, 0)$.

Since $\Delta_1 = \frac{1}{2} > 0$, the present solution is not optimal.

Step 8: Find the incoming and outgoing vectors.

Since $\Delta_1 = \frac{1}{2}$ is the maximum positive value, Y_1 is the incoming vector. In the Y_1 -column, only the first entry is positive: $\frac{x_{B1}}{y_{11}} = \frac{10}{5/2} = 4$. The remaining entries are negative: $-\frac{1}{2} < 0$, $-\frac{5}{2} < 0$.

Hence, Y_4 is the outgoing vector. The pivot element is $y_{11} = \frac{5}{2}$.

Step 9: Perform the second pivot row operations.

First divide the first row by $\frac{5}{2}$: $R''_1 = \frac{R'_1}{5/2} = \frac{2}{5} R'_1 = \frac{2}{5} [10, \frac{5}{2}, 0, 2, 1, \frac{1}{4}, 0] = [4, 1, 0, \frac{4}{5}, \frac{2}{5}, \frac{1}{10}, 0]$. Hence,

$R_1'' = [4, 1, 0, \frac{4}{5}, \frac{2}{5}, \frac{1}{10}, 0]$. Now eliminate $-\frac{1}{2}$ from the Y_1 -column of the second row.

$$R_2'' = R_2' + \frac{1}{2}R_1'' = [3, -\frac{1}{2}, 1, 0, 0, \frac{1}{4}, 0] + \frac{1}{2}[4, 1, 0, \frac{4}{5}, \frac{2}{5}, \frac{1}{10}, 0] = [5, 0, 1, \frac{2}{5}, \frac{1}{5}, \frac{3}{10}, 0].$$

Thus, $R_2'' = [5, 0, 1, \frac{2}{5}, \frac{1}{5}, \frac{3}{10}, 0]$. Now eliminate $-\frac{5}{2}$ from the Y_1 -column of the third row: $R_3'' = R_3' + \frac{5}{2}R_1'' = [1, -\frac{5}{2}, 0, 8, 0, -\frac{3}{4}, 1] + \frac{5}{2}[4, 1, 0, \frac{4}{5}, \frac{2}{5}, \frac{1}{10}, 0] = [11, 0, 0, 10, 1, -\frac{1}{2}, 1]$. Therefore, $R_3'' = [11, 0, 0, 10, 1, -\frac{1}{2}, 1]$. The new basis is $B = \{Y_1, Y_2, Y_6\}$, and $c_B = (-1, 3, 0)$. The new objective value is $Z' = (-1)(4) + (3)(5) + (0)(11) = -4 + 15 = 11$.

Third Simplex Table

B	c_B	c_j	-1	3	-2	0	0	0	Status
		x_B	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	
Y_1	-1	4	1	0	$\frac{4}{5}$	$\frac{2}{5}$	$\frac{1}{10}$	0	—
Y_2	3	5	0	1	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{3}{10}$	0	—
Y_6	0	11	0	0	10	1	$-\frac{1}{2}$	1	—
$Z' = c_B^T x_B = 11$		Δ_j	0	0	$-\frac{12}{5}$	$-\frac{1}{5}$	$-\frac{4}{5}$	0	Optimal

Step 10: Final optimality test.

Since Y_1, Y_2, Y_6 are basic columns, $\Delta_1 = \Delta_2 = \Delta_6 = 0$. For Y_3 ,

$$\Delta_3 = -2 - (-1, 3, 0) \begin{pmatrix} \frac{4}{5} \\ \frac{2}{5} \\ \frac{1}{10} \end{pmatrix} = -2 - (-\frac{4}{5} + \frac{6}{5}) = -2 - \frac{2}{5} = -\frac{12}{5}. \text{ For } Y_4, \Delta_4 = 0 - (-1, 3, 0) \begin{pmatrix} \frac{2}{5} \\ \frac{1}{5} \\ \frac{1}{10} \end{pmatrix} =$$

$$-(-\frac{2}{5} + \frac{3}{5}) = -\frac{1}{5}. \text{ For } Y_5, \Delta_5 = 0 - (-1, 3, 0) \begin{pmatrix} \frac{1}{10} \\ \frac{3}{10} \\ -\frac{1}{2} \end{pmatrix} = -(-\frac{1}{10} + \frac{9}{10}) = -\frac{4}{5}. \text{ Therefore, } \Delta_1 =$$

0, $\Delta_2 = 0$, $\Delta_3 = -\frac{12}{5}$, $\Delta_4 = -\frac{1}{5}$, $\Delta_5 = -\frac{4}{5}$, $\Delta_6 = 0$. Since $\Delta_j \leq 0$ for all j , the present solution is optimal.

Step 11: Write the optimal solution.

From the final simplex table, the basic variables are $x_1 = 4$, $x_2 = 5$, $x_6 = 11$. The non-basic variables are $x_3 = x_4 = x_5 = 0$. Therefore, $x_1 = 4$, $x_2 = 5$, $x_3 = 0$. The maximum value of Z' is $Z'_{\max} = -4 + 15 = 11$. But $Z' = -Z$.

Therefore, $-Z = 11$, so that $Z_{\min} = -11$. Hence, the required optimal solution is $x_1 = 4$, $x_2 = 5$, $x_3 = 0$, $Z_{\min} = -11$. Moreover, the reduced costs corresponding to all non-basic variables are $\Delta_3 = -\frac{12}{5} < 0$, $\Delta_4 = -\frac{1}{5} < 0$, $\Delta_5 = -\frac{4}{5} < 0$. Hence, the obtained solution is the **unique optimal solution**.

The successive simplex computations can be represented conveniently in the following single table.

Table 11: Single combined simplex table for all iterations

B	c_B	c_j X_B	-1 Y_1	3 Y_2	-2 Y_3	0 $Y_4(\beta_1)$	0 $Y_5(\beta_2)$	0 $Y_6(\beta_3)$	Minimum ratio Entering column
Y_4	0	7	3	-1	2	1	0	0	Neg.
Y_5	0	12	-2	$\boxed{4}$	0	0	1	0	$\frac{12}{4} = 3$ (minimum)
Y_6	0	10	-4	3	8	0	0	1	$\frac{10}{3}$
$Z' = 0$		Δ_j	-1	3	-2	0	0	0	x_2 enters
Y_4	0	10	$\frac{5}{2}$	0	2	1	$\frac{1}{4}$	0	$\frac{10}{5/2} = 4$ (minimum)
Y_2	3	3	$-\frac{1}{2}$	1	0	0	$\frac{1}{4}$	0	Neg.
Y_6	0	1	$-\frac{5}{2}$	0	8	0	$-\frac{3}{4}$	1	Neg.
$Z' = 9$		Δ_j	$\frac{1}{2}$	0	-2	0	$-\frac{3}{4}$	0	x_1 enters
Y_1	-1	4	1	0	$\frac{4}{5}$	$\frac{2}{5}$	$\frac{1}{10}$	0	—
Y_2	3	5	0	1	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{3}{10}$	0	—
Y_6	0	11	0	0	10	1	$-\frac{1}{2}$	1	—
$Z' = 11$		Δ_j	0	0	$-\frac{12}{5}$	$-\frac{1}{5}$	$-\frac{4}{5}$	0	Optimal

First Pivot: From the first part of the table, $\Delta_2 = 3$ is the maximum positive value of Δ_j . Therefore, Y_2 is the **incoming vector**. Applying the minimum ratio test, $\min \left\{ \frac{12}{4}, \frac{10}{3} \right\} = \frac{12}{4} = 3$. Hence, Y_5 is the **outgoing vector**, and $y_{22} = 4$ is the key element. After performing the required row operations, we obtain the second iteration shown in the same table.

Second Pivot: For the second iteration, $\Delta_1 = \frac{1}{2}$ is the maximum positive value of Δ_j . Therefore, Y_1 is the **incoming vector**. By the minimum ratio rule, $\frac{10}{5/2} = 4$. Hence, Y_4 is the **outgoing vector**, and $y_{11} = \frac{5}{2}$ is the key element.

After performing the required row operations, we obtain the final part of the table. Since $\Delta_3 = -\frac{12}{5}$, $\Delta_4 = -\frac{1}{5}$, $\Delta_5 = -\frac{4}{5}$, all the Δ_j corresponding to the non-basic variables are negative. Therefore, the solution is optimal. Thus, $x_1 = 4$, $x_2 = 5$, $x_3 = 0$. Also, $Z' = 11$. Since $Z' = -Z$, we obtain $\boxed{\min Z = -11}$. Hence, the optimal solution is

$$\boxed{x_1 = 4, \quad x_2 = 5, \quad x_3 = 0, \quad \min Z = -11}.$$

Example 4:

Solve by the simplex method:

$$\begin{aligned} \text{Maximize} \quad & Z = x_1 - x_2 + 3x_3, \\ \text{subject to} \quad & x_1 + x_2 + x_3 \leq 10, \\ & 2x_1 - x_3 \leq 2, \\ & 2x_1 - 2x_2 + 3x_3 \leq 0, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Solution: Step 1: Preparation.

The given problem is already a maximization problem and all the right-hand side constants are non-negative.

Step 2: Convert the constraints into standard form.

Introduce the slack variables $x_4, x_5, x_6 \geq 0$. Then the constraints become

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 &= 10, \\ 2x_1 - x_3 + x_5 &= 2, \\ 2x_1 - 2x_2 + 3x_3 + x_6 &= 0. \end{aligned}$$

The objective function is $Z = x_1 - x_2 + 3x_3 + 0x_4 + 0x_5 + 0x_6$. Therefore, $c_j = (1, -1, 3, 0, 0, 0)$.

Step 3: Find the initial basic feasible solution.

Initially, the slack variables x_4, x_5, x_6 are basic variables. Setting the non-basic variables equal to zero, $x_1 = x_2 = x_3 = 0$, we obtain $x_4 = 10$, $x_5 = 2$, $x_6 = 0$. Hence, the initial B.F.S. is $(x_1, x_2, x_3, x_4, x_5, x_6) = (0, 0, 0, 10, 2, 0)$, with $Z = 0$. Since the basic variable $x_6 = 0$, the initial basic feasible solution is **degenerate**. The initial basis is $B = \{Y_4, Y_5, Y_6\}$, and $c_B = (0, 0, 0)$.

First Simplex Table

B	c_B	c_j	1	-1	3	0	0	0	Min. ratio $x_B/Y_3, \quad y_{i3} > 0$
		x_B	Y_1	Y_2	Y_3	$Y_4(\beta_1)$	$Y_5(\beta_2)$	$Y_6(\beta_3)$	
Y_4	0	10	1	1	1	1	0	0	$\frac{10}{1} = 10$
Y_5	0	2	2	0	-1	0	1	0	Neg.
Y_6	0	0	2	-2	3	0	0	1	$\frac{0}{3} = 0$ (min)
$Z = c_B^T x_B = 0$		Δ_j	1	-1	3 $\uparrow$	0	0	0	x_3 enters

Step 4: Optimality test for the first simplex table.

Using $\Delta_j = c_j - c_B^T Y_j$, we obtain $\Delta_1 = 1 - (0, 0, 0) \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = 1$, $\Delta_2 = -1 - (0, 0, 0) \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} = -1$, and

$\Delta_3 = 3 - (0, 0, 0) \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} = 3$. Therefore, $\Delta_1 = 1$, $\Delta_2 = -1$, $\Delta_3 = 3$. Since some $\Delta_j > 0$,

the present solution is not optimal.

Step 5: Find the incoming and outgoing vectors.

Since $\Delta_3 = 3$ is the maximum positive value of Δ_j , Y_3 is the incoming vector. Now apply the minimum ratio test to the Y_3 -column. For the first row, $\frac{x_{B1}}{y_{13}} = \frac{10}{1} = 10$. The second row is not considered since $y_{23} = -1 < 0$. For the third row, $\frac{x_{B3}}{y_{33}} = \frac{0}{3} = 0$. Hence, $\min\{10, 0\} = 0$. Therefore, Y_6 is the outgoing vector. The pivot element is $y_{33} = 3$. Since the minimum ratio is zero, this pivot is a **degenerate pivot**.

Step 6: Perform the first pivot row operations.

The original rows are $R_1 = [10, 1, 1, 1, 1, 0, 0]$, $R_2 = [2, 2, 0, -1, 0, 1, 0]$, and $R_3 = [0, 2, -2, 3, 0, 0, 1]$. Since the pivot element is 3, divide the third row by 3:

$R'_3 = \frac{1}{3}R_3 = \frac{1}{3}[0, 2, -2, 3, 0, 0, 1] = [0, \frac{2}{3}, -\frac{2}{3}, 1, 0, 0, \frac{1}{3}]$. Therefore, $R'_3 = [0, \frac{2}{3}, -\frac{2}{3}, 1, 0, 0, \frac{1}{3}]$.

Now eliminate the entry 1 from the Y_3 -column of the first row: $R'_1 = R_1 - R'_3 = [10, 1, 1, 1, 1, 0, 0] - [0, \frac{2}{3}, -\frac{2}{3}, 1, 0, 0, \frac{1}{3}] = [10, \frac{1}{3}, \frac{5}{3}, 0, 1, 0, -\frac{1}{3}]$. Thus, $R'_1 = [10, \frac{1}{3}, \frac{5}{3}, 0, 1, 0, -\frac{1}{3}]$.

The coefficient of Y_3 in the second row is -1. Therefore, $R'_2 = R_2 + R'_3$. Hence, $R'_2 = [2, 2, 0, -1, 0, 1, 0] + [0, \frac{2}{3}, -\frac{2}{3}, 1, 0, 0, \frac{1}{3}] = [2, \frac{8}{3}, -\frac{2}{3}, 0, 0, 1, \frac{1}{3}]$. Therefore, $R'_2 = [2, \frac{8}{3}, -\frac{2}{3}, 0, 0, 1, \frac{1}{3}]$. The new basis is

$B = \{Y_4, Y_5, Y_3\}$, and $c_B = (0, 0, 3)$. The new objective value is $Z = (0)(10) + (0)(2) + (3)(0) = 0$. For the reduced-cost row, $R'_\Delta = R_\Delta - 3R'_3 = [1, -1, 3, 0, 0, 0] - 3[0, \frac{2}{3}, -\frac{2}{3}, 1, 0, 0, \frac{1}{3}] = [-1, 1, 0, 0, 0, -1]$. Therefore, $4(\Delta_1, \Delta_2, \Delta_3, \Delta_4, \Delta_5, \Delta_6) = (-1, 1, 0, 0, 0, -1)$.

Second Simplex Table

B	c_B	c_j	1	-1	3	0	0	0	Min. ratio $x_B/Y_2, \quad y_{i2} > 0$
		x_B	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	
Y_4	0	10	$\frac{1}{3}$	$\frac{5}{3}$	0	1	0	$-\frac{1}{3}$	$\frac{10}{5/3} = 6$ (min)
Y_5	0	2	$\frac{8}{3}$	$-\frac{2}{3}$	0	0	1	$\frac{1}{3}$	Neg.
Y_3	3	0	$\frac{2}{3}$	$-\frac{2}{3}$	1	0	0	$\frac{1}{3}$	Neg.
$Z = c_B^T x_B = 0$		Δ_j	-1	1 ↑	0	0	0	-1	x_2 enters

Step 7: Optimality test for the second simplex table.

Since Y_3, Y_4, Y_5 are basic columns, $\Delta_3 = \Delta_4 = \Delta_5 = 0$. For Y_1 , $\Delta_1 = 1 - (0, 0, 3) \begin{pmatrix} 1 \\ \frac{1}{3} \\ \frac{8}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{pmatrix} = 1 - 2 = -1$.

For Y_2 , $\Delta_2 = -1 - (0, 0, 3) \begin{pmatrix} \frac{5}{3} \\ \frac{2}{3} \\ -\frac{2}{3} \\ -\frac{2}{3} \\ \frac{1}{3} \end{pmatrix} = -1 - (-2) = 1$. For Y_6 , $\Delta_6 = 0 - (0, 0, 3) \begin{pmatrix} -\frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix} = -1$.

Therefore, $(\Delta_1, \Delta_2, \Delta_3, \Delta_4, \Delta_5, \Delta_6) = (-1, 1, 0, 0, 0, -1)$. Since $\Delta_2 = 1 > 0$, the present solution is not optimal.

Step 8: Find the incoming and outgoing vectors.

Since $\Delta_2 = 1$ is the maximum positive value, $\boxed{Y_2 \text{ is the incoming vector.}}$ For the Y_2 -column, only the first entry is positive. Thus, $\frac{x_{B1}}{y_{12}} = \frac{10}{5/3} = 10 \left(\frac{3}{5}\right) = 6$. The other two entries are negative:

$-\frac{2}{3} < 0, \quad -\frac{2}{3} < 0$. Therefore, $\boxed{Y_4 \text{ is the outgoing vector.}}$ The pivot element is $\boxed{y_{12} = \frac{5}{3}}$.

Step 9: Perform the second pivot row operations.

First divide the first row by $\frac{5}{3}$:

$$R_1'' = \frac{R_1'}{5/3} = \frac{3}{5} R_1' = \frac{3}{5} [10, \frac{1}{3}, \frac{5}{3}, 0, 1, 0, -\frac{1}{3}] = [6, \frac{1}{5}, 1, 0, \frac{3}{5}, 0, -\frac{1}{5}]. \text{ Thus, } R_1'' = [6, \frac{1}{5}, 1, 0, \frac{3}{5}, 0, -\frac{1}{5}].$$

Now eliminate $-\frac{2}{3}$ from the Y_2 -column of the second row:

$$R_2'' = R_2' + \frac{2}{3} R_1'' = \left[2, \frac{8}{3}, -\frac{2}{3}, 0, 0, 1, \frac{1}{3}\right] + \frac{2}{3} \left[6, \frac{1}{5}, 1, 0, \frac{3}{5}, 0, -\frac{1}{5}\right] = \left[6, \frac{14}{5}, 0, 0, \frac{2}{5}, 1, \frac{1}{5}\right].$$

Therefore, $R_2'' = [6, \frac{14}{5}, 0, 0, \frac{2}{5}, 1, \frac{1}{5}]$.

Similarly, eliminate $-\frac{2}{3}$ from the third row:

$$R_3'' = R_3' + \frac{2}{3} R_1'' = \left[0, \frac{2}{3}, -\frac{2}{3}, 1, 0, 0, \frac{1}{3}\right] + \frac{2}{3} \left[6, \frac{1}{5}, 1, 0, \frac{3}{5}, 0, -\frac{1}{5}\right] = \left[4, \frac{4}{5}, 0, 1, \frac{2}{5}, 0, \frac{1}{5}\right]. \text{ Hence, } R_3'' = \left[4, \frac{4}{5}, 0, 1, \frac{2}{5}, 0, \frac{1}{5}\right].$$

The new basis is $\boxed{B = \{Y_2, Y_5, Y_3\}}$, and $c_B = (-1, 0, 3)$. The new objective value is

$Z = (-1)(6) + (0)(6) + (3)(4) = -6 + 12 = 6$. For the reduced-cost row,

$$R''_{\Delta} = R'_{\Delta} - R''_1 = [-1, 1, 0, 0, 0, -1] - \left[\frac{1}{5}, 1, 0, \frac{3}{5}, 0, -\frac{1}{5} \right] = \left[-\frac{6}{5}, 0, 0, -\frac{3}{5}, 0, -\frac{4}{5} \right].$$

Therefore, $(\Delta_1, \Delta_2, \Delta_3, \Delta_4, \Delta_5, \Delta_6) = \left(-\frac{6}{5}, 0, 0, -\frac{3}{5}, 0, -\frac{4}{5} \right)$.

Third Simplex Table

B	c_B	c_j	1	-1	3	0	0	0	Status
		x_B	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	
Y_2	-1	6	$\frac{1}{5}$	1	0	$\frac{3}{5}$	0	$-\frac{1}{5}$	—
Y_5	0	6	$\frac{14}{5}$	0	0	$\frac{2}{5}$	1	$\frac{1}{5}$	—
Y_3	3	4	$\frac{4}{5}$	0	1	$\frac{2}{5}$	0	$\frac{1}{5}$	—
$Z = c_B^T x_B = 6$	Δ_j		$-\frac{6}{5}$	0	0	$-\frac{3}{5}$	0	$-\frac{4}{5}$	Optimal

Step 10: Final optimality test.

Since Y_2, Y_3, Y_5 are basic columns, $\Delta_2 = \Delta_3 = \Delta_5 = 0$. For Y_1 ,

$$\Delta_1 = 1 - (-1, 0, 3) \begin{pmatrix} \frac{1}{5} \\ \frac{14}{5} \\ \frac{4}{5} \end{pmatrix} = 1 - \left(-\frac{1}{5} + \frac{12}{5} \right) = 1 - \frac{11}{5} = -\frac{6}{5}.$$

For Y_4 , $\Delta_4 = 0 - (-1, 0, 3) \begin{pmatrix} \frac{3}{5} \\ \frac{2}{5} \\ \frac{2}{5} \end{pmatrix} = -\left(-\frac{3}{5} + \frac{6}{5} \right) = -\frac{3}{5}$. For Y_6 , $\Delta_6 = 0 - (-1, 0, 3) \begin{pmatrix} -\frac{1}{5} \\ \frac{1}{5} \\ \frac{1}{5} \end{pmatrix} = -\left(\frac{1}{5} + \frac{3}{5} \right) = -\frac{4}{5}$.

Therefore, $\Delta_1 = -\frac{6}{5}$, $\Delta_2 = 0$, $\Delta_3 = 0$, $\Delta_4 = -\frac{3}{5}$, $\Delta_5 = 0$, $\Delta_6 = -\frac{4}{5}$. Since $\Delta_j \leq 0$ for all j , the present solution is optimal.

Step 11: Write the optimal solution.

From the final simplex table, the basic variables are $x_2 = 6$, $x_5 = 6$, $x_3 = 4$. Therefore, the non-basic variables are $x_1 = x_4 = x_6 = 0$. Hence, $x_1 = 0$, $x_2 = 6$, $x_3 = 4$

The maximum value of the objective function is

$Z_{\max} = x_1 - x_2 + 3x_3 = 0 - 6 + 3(4) = -6 + 12 = 6$. Thus, the required optimal solution is

$$x_1 = 0, \quad x_2 = 6, \quad x_3 = 4, \quad Z_{\max} = 6.$$

Moreover, the reduced costs corresponding to the non-basic variables are

$$\Delta_1 = -\frac{6}{5} < 0, \quad \Delta_4 = -\frac{3}{5} < 0, \quad \Delta_6 = -\frac{4}{5} < 0.$$

Hence, the obtained solution is the **unique optimal solution**.

Table 12: Single combined simplex table

B	c_B	x_B	1	-1	3	0	0	0	Minimum ratio
			Y_1	Y_2	Y_3	$Y_4(\beta_1)$	$Y_5(\beta_2)$	$Y_6(\beta_3)$	Entering column
Y_4	0	10	1	1	1	1	0	0	$10/1 = 10$
Y_5	0	2	2	0	-1	0	1	0	Not eligible
Y_6	0	0	2	-2	3	0	0	1	$0/3 = 0$ (minimum)
$Z = 0$		Δ_j	1	-1	3	0	0	0	x_3 enters
Y_4	0	10	1/3	5/3	0	1	0	-1/3	$10/(5/3) = 6$ (minimum)
Y_5	0	2	8/3	-2/3	0	0	1	1/3	Not eligible
Y_3	3	0	2/3	-2/3	1	0	0	1/3	Not eligible
$Z = 0$		Δ_j	-1	1	0	0	0	-1	x_2 enters
Y_2	-1	6	1/5	1	0	3/5	0	-1/5	—
Y_5	0	6	14/5	0	0	2/5	1	1/5	—
Y_3	3	4	4/5	0	1	2/5	0	1/5	—
$Z = 6$		Δ_j	-6/5	0	0	-3/5	0	-4/5	Optimal

First pivot: a degenerate pivot. Initially, $\Delta_3 = 3$ is the largest positive reduced cost, so x_3 enters. Only positive entries in the x_3 column are eligible:

$$\frac{10}{1} = 10, \quad \frac{0}{3} = 0.$$

The minimum ratio is zero, so x_6 leaves and the pivot is 3. Perform

$$R_3 \leftarrow \frac{1}{3}R_3, \quad R_1 \leftarrow R_1 - R_3, \quad R_2 \leftarrow R_2 + R_3,$$

where the last two operations use the normalized third row. Although x_3 enters the basis, its new value is zero. Hence the basis changes but the objective value remains $Z = 0$. This is called a *degenerate pivot*.

Second pivot. In the second block, $\Delta_2 = 1 > 0$, so x_2 enters. The only positive entry in its column is $5/3$ in the x_4 row. Thus

$$\frac{10}{5/3} = 6,$$

so x_4 leaves and the pivot is $5/3$. Perform

$$R_1 \leftarrow \frac{3}{5}R_1, \quad R_2 \leftarrow R_2 + \frac{2}{3}R_1, \quad R_3 \leftarrow R_3 + \frac{2}{3}R_1,$$

where the last two operations use the normalized first row.

In the final block, every reduced cost satisfies $\Delta_j \leq 0$; therefore the solution is optimal. Reading the basic variables gives

$$\boxed{x_1 = 0, \quad x_2 = 6, \quad x_3 = 4, \quad Z_{\max} = 6.}$$

The slack variables are $x_4 = 0$, $x_5 = 6$ and $x_6 = 0$. The non-basic variables x_1, x_4, x_6 have strictly negative reduced costs, so the optimal solution is unique.

Example 5:

Solve by the simplex method:

$$\begin{aligned} \text{Maximize} \quad & Z = 4x_1 + 7x_2, \\ \text{subject to} \quad & 2x_1 + x_2 \leq 1000, \\ & x_1 + x_2 \leq 600, \\ & -x_1 - 2x_2 \geq -1000, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Solution: Step 1: Preparation.

The given problem is already a maximization problem. However, the right-hand side of the third constraint is negative. Therefore, multiply the third constraint by -1 . Thus, $-x_1 - 2x_2 \geq -1000$ becomes $x_1 + 2x_2 \leq 1000$. Hence, the equivalent L.P. problem is

$$\begin{aligned} \text{Maximize} \quad & Z = 4x_1 + 7x_2, \\ \text{subject to} \quad & 2x_1 + x_2 \leq 1000, \\ & x_1 + x_2 \leq 600, \\ & x_1 + 2x_2 \leq 1000, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Step 2: Convert the constraints into standard form.

Introduce the slack variables $x_3, x_4, x_5 \geq 0$. Then the constraints become

$$\begin{aligned} 2x_1 + x_2 + x_3 &= 1000, \\ x_1 + x_2 + x_4 &= 600, \\ x_1 + 2x_2 + x_5 &= 1000. \end{aligned}$$

The objective function is $Z = 4x_1 + 7x_2 + 0x_3 + 0x_4 + 0x_5$. Therefore, $c_j = (4, 7, 0, 0, 0)$.

Step 3: Find the initial basic feasible solution.

Initially, the slack variables x_3, x_4, x_5 are basic variables. Setting the non-basic variables equal to zero, $x_1 = x_2 = 0$, we obtain $x_3 = 1000$, $x_4 = 600$, $x_5 = 1000$. Hence, the initial B.F.S. is $(x_1, x_2, x_3, x_4, x_5) = (0, 0, 1000, 600, 1000)$, with $Z = 0$. The initial basis is $B = \{Y_3, Y_4, Y_5\}$, and $c_B = (0, 0, 0)$.

First Simplex Table

B	c_B	c_j	4	7	0	0	0	Min. ratio $x_B/Y_2, \quad y_{i2} > 0$
		x_B	Y_1	Y_2	$Y_3(\beta_1)$	$Y_4(\beta_2)$	$Y_5(\beta_3)$	
Y_3	0	1000	2	1	1	0	0	$\frac{1000}{1} = 1000$
Y_4	0	600	1	1	0	1	0	$\frac{600}{1} = 600$
Y_5	0	1000	1	2	0	0	1	$\frac{1000}{2} = 500$ (min)
$Z = c_B^T x_B = 0$		Δ_j	4	7 $\uparrow$	0	0	0	x_2 enters

Step 4: Optimality test for the first simplex table.

Using $\Delta_j = c_j - c_B^T Y_j$, we obtain $\Delta_1 = 4 - (0, 0, 0) \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = 4$, and $\Delta_2 = 7 - (0, 0, 0) \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 7$. Thus, $\Delta_1 = 4$, $\Delta_2 = 7$. Since some $\Delta_j > 0$, the present solution is not optimal.

Step 5: Find the incoming and outgoing vectors.

Since $\Delta_2 = 7$ is the maximum positive value of Δ_j , Y_2 is the incoming vector. Now apply the minimum ratio test to the Y_2 -column. For the first row, $\frac{x_{B1}}{y_{12}} = \frac{1000}{1} = 1000$. For the second row, $\frac{x_{B2}}{y_{22}} = \frac{600}{1} = 600$. For the third row, $\frac{x_{B3}}{y_{32}} = \frac{1000}{2} = 500$. Hence, $\min\{1000, 600, 500\} = 500$. Therefore, Y_5 is the outgoing vector. The pivot element is $y_{32} = 2$.

Step 6: Perform the first pivot row operations.

The original rows are $R_1 = [1000, 2, 1, 1, 0, 0]$, $R_2 = [600, 1, 1, 0, 1, 0]$, and $R_3 = [1000, 1, 2, 0, 0, 1]$. Since the pivot element is 2, divide the third row by 2:

$$R'_3 = \frac{1}{2}R_3 = \frac{1}{2}[1000, 1, 2, 0, 0, 1] = [500, \frac{1}{2}, 1, 0, 0, \frac{1}{2}]. \text{ Therefore, } R'_3 = [500, \frac{1}{2}, 1, 0, 0, \frac{1}{2}].$$

Now eliminate the entry 1 from the Y_2 -column of the first row:

$$R'_1 = R_1 - R'_3 = [1000, 2, 1, 1, 0, 0] - \left[500, \frac{1}{2}, 1, 0, 0, \frac{1}{2}\right] = \left[500, \frac{3}{2}, 0, 1, 0, -\frac{1}{2}\right].$$

Thus, $R'_1 = [500, \frac{3}{2}, 0, 1, 0, -\frac{1}{2}]$. Similarly, $R'_2 = R_2 - R'_3 = [600, 1, 1, 0, 1, 0] - [500, \frac{1}{2}, 1, 0, 0, \frac{1}{2}] = [100, \frac{1}{2}, 0, 0, 1, -\frac{1}{2}]$. Therefore, $R'_2 = [100, \frac{1}{2}, 0, 0, 1, -\frac{1}{2}]$. The new basis is $B = \{Y_3, Y_4, Y_2\}$, and

$c_B = (0, 0, 7)$. The new objective value is $Z = (0)(500) + (0)(100) + (7)(500) = 3500$. For the reduced-cost row,

$R'_\Delta = R_\Delta - 7R'_3 = [4, 7, 0, 0, 0] - 7\left[\frac{1}{2}, 1, 0, 0, \frac{1}{2}\right] = \left[\frac{1}{2}, 0, 0, 0, -\frac{7}{2}\right]$. Thus, $(\Delta_1, \Delta_2, \Delta_3, \Delta_4, \Delta_5) = \left(\frac{1}{2}, 0, 0, 0, -\frac{7}{2}\right)$.

Second Simplex Table

B	c_B	c_j	4	7	0	0	0	Min. ratio $x_B/Y_1, \quad y_{i1} > 0$
		x_B	Y_1	Y_2	Y_3	Y_4	Y_5	
Y_3	0	500	$\frac{3}{2}$	0	1	0	$-\frac{1}{2}$	$\frac{500}{3/2} = \frac{1000}{3}$
Y_4	0	100	$\frac{1}{2}$	0	0	1	$-\frac{1}{2}$	$\frac{100}{1/2} = 200$ (min)
Y_2	7	500	$\frac{1}{2}$	1	0	0	$\frac{1}{2}$	$\frac{500}{1/2} = 1000$
$Z = c_B^T x_B = 3500$		Δ_j	$\frac{1}{2}$ $\uparrow$	0	0	0	$-\frac{7}{2}$	x_1 enters

Step 7: Optimality test for the second simplex table.

Since Y_2, Y_3, Y_4 are basic columns, $\Delta_2 = \Delta_3 = \Delta_4 = 0$. For Y_1 , $\Delta_1 = 4 - (0, 0, 7) \begin{pmatrix} \frac{3}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = 4 - \frac{7}{2} = \frac{1}{2}$.

For Y_5 , $\Delta_5 = 0 - (0, 0, 7) \begin{pmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = -\frac{7}{2}$. Therefore, $(\Delta_1, \Delta_2, \Delta_3, \Delta_4, \Delta_5) = \left(\frac{1}{2}, 0, 0, 0, -\frac{7}{2}\right)$. Since

$\Delta_1 = \frac{1}{2} > 0$, the present solution is not optimal.

Step 8: Find the incoming and outgoing vectors.

Since $\Delta_1 = \frac{1}{2}$ is the maximum positive value, Y_1 is the incoming vector. Now apply the minimum ratio test to the Y_1 -column.

For the first row, $\frac{x_{B1}}{y_{11}} = \frac{500}{3/2} = \frac{1000}{3}$. For the second row, $\frac{x_{B2}}{y_{21}} = \frac{100}{1/2} = 200$. For the third row, $\frac{x_{B3}}{y_{31}} = \frac{500}{1/2} = 1000$. Hence, $\min\left\{\frac{1000}{3}, 200, 1000\right\} = 200$. Therefore, Y_4 is the outgoing vector. The

pivot element is $y_{21} = \frac{1}{2}$.

Step 9: Perform the second pivot row operations.

First divide the second row by $\frac{1}{2}$:

$$R''_2 = \frac{R'_2}{1/2} = 2R'_2 = 2\left[100, \frac{1}{2}, 0, 0, 1, -\frac{1}{2}\right] = [200, 1, 0, 0, 2, -1].$$

Thus, $R''_2 = [200, 1, 0, 0, 2, -1]$. Now eliminate $\frac{3}{2}$ from the Y_1 -column of the first row:

$$R''_1 = R'_1 - \frac{3}{2}R''_2 = \left[500, \frac{3}{2}, 0, 1, 0, -\frac{1}{2}\right] - \frac{3}{2}[200, 1, 0, 0, 2, -1] = [200, 0, 0, 1, -3, 1].$$

Therefore, $R'_1 = [200, 0, 0, 1, -3, 1]$. Now eliminate $\frac{1}{2}$ from the Y_1 -column of the third row:

$$R''_3 = R'_3 - \frac{1}{2}R''_2 = \left[500, \frac{1}{2}, 1, 0, 0, \frac{1}{2}\right] - \frac{1}{2}[200, 1, 0, 0, 2, -1] = [400, 0, 1, 0, -1, 1].$$

Hence, $R''_3 = [400, 0, 1, 0, -1, 1]$. The new basis is $B = \{Y_3, Y_1, Y_2\}$, and $c_B = (0, 4, 7)$. The new objective value is $Z = (0)(200) + (4)(200) + (7)(400) = 800 + 2800 = 3600$. For the reduced-cost row, $R''_\Delta = R'_\Delta - \frac{1}{2}R''_2 = \left[\frac{1}{2}, 0, 0, 0, -\frac{7}{2}\right] - \frac{1}{2}[1, 0, 0, 2, -1] = [0, 0, 0, -1, -3]$.

Therefore, $(\Delta_1, \Delta_2, \Delta_3, \Delta_4, \Delta_5) = (0, 0, 0, -1, -3)$.

Third Simplex Table

B	c_B	c_j	4	7	0	0	0	Status
		x_B	Y_1	Y_2	Y_3	Y_4	Y_5	
Y_3	0	200	0	0	1	-3	1	—
Y_1	4	200	1	0	0	2	-1	—
Y_2	7	400	0	1	0	-1	1	—
$Z = c_B^T x_B = 3600$		Δ_j	0	0	0	-1	-3	Optimal

Step 10: Final optimality test.

Since Y_1, Y_2, Y_3 are basic columns, $\Delta_1 = \Delta_2 = \Delta_3 = 0$. For Y_4 , $\Delta_4 = 0 - (0, 4, 7) \begin{pmatrix} -3 \\ 2 \\ -1 \end{pmatrix} = -(8 - 7) = -1$. For Y_5 , $\Delta_5 = 0 - (0, 4, 7) \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = -(-4 + 7) = -3$. Thus, $\Delta_1 = 0, \Delta_2 = 0, \Delta_3 = 0, \Delta_4 = -1, \Delta_5 = -3$. Since $\Delta_j \leq 0$ for all j , the present solution is optimal.

Step 11: Write the optimal solution.

From the final simplex table, the basic variables are $x_3 = 200$, $x_1 = 200$, $x_2 = 400$. Thus, $x_1 = 200$, $x_2 = 400$. The non-basic slack variables are $x_4 = x_5 = 0$. Therefore, $x_1 = 200$, $x_2 = 400$.

The maximum value of the objective function is $Z_{\max} = 4(200) + 7(400) = 800 + 2800 = 3600$. Hence, $x_1 = 200$, $x_2 = 400$, $Z_{\max} = 3600$.

Moreover, the reduced costs corresponding to the non-basic variables are $\Delta_4 = -1 < 0$, $\Delta_5 = -3 < 0$. Hence, the obtained solution is the **unique optimal solution**.

Table 13: Single combined simplex table

B	c_B	x_B	4 x_1	7 x_2	0 x_3	0 x_4	0 x_5	Minimum ratio Entering column
x_3	0	1000	2	1	1	0	0	1000/1 = 1000
x_4	0	600	1	1	0	1	0	600/1 = 600
x_5	0	1000	1	2	0	0	1	1000/2 = 500 (minimum)
$Z = 0$		Δ_j	4	7	0	0	0	x_2 enters
x_3	0	500	3/2	0	1	0	-1/2	500/(3/2) = 1000/3
x_4	0	100	1/2	0	0	1	-1/2	100/(1/2) = 200 (minimum)
x_2	7	500	1/2	1	0	0	1/2	500/(1/2) = 1000
$Z = 3500$		Δ_j	1/2	0	0	0	-7/2	x_1 enters
x_3	0	200	0	0	1	-3	1	—
x_1	4	200	1	0	0	2	-1	—
x_2	7	400	0	1	0	-1	1	—
$Z = 3600$		Δ_j	0	0	0	-1	-3	Optimal

First pivot. Initially, the largest positive reduced cost is $\Delta_2 = 7$, so x_2 enters. The positive ratios are

$$\frac{1000}{1} = 1000, \quad \frac{600}{1} = 600, \quad \frac{1000}{2} = 500.$$

The minimum ratio is 500, so x_5 leaves and the pivot is 2. Perform

$$R_3 \leftarrow \frac{1}{2}R_3, \quad R_1 \leftarrow R_1 - R_3, \quad R_2 \leftarrow R_2 - R_3,$$

where the last two operations use the normalized third row. The new basis is $\{x_3, x_4, x_2\}$ and $Z = 3500$.

Second pivot. In the second block, $\Delta_1 = 1/2 > 0$, so x_1 enters. The ratios are

$$\frac{500}{3/2} = \frac{1000}{3}, \quad \frac{100}{1/2} = 200, \quad \frac{500}{1/2} = 1000.$$

The minimum ratio is 200, so x_4 leaves and the pivot is $1/2$. Perform

$$R_2 \leftarrow 2R_2, \quad R_1 \leftarrow R_1 - \frac{3}{2}R_2, \quad R_3 \leftarrow R_3 - \frac{1}{2}R_2,$$

where the last two operations use the normalized second row.

In the final block, every reduced cost satisfies $\Delta_j \leq 0$, so the solution is optimal. Hence

$$\boxed{x_1 = 200, \quad x_2 = 400, \quad Z_{\max} = 3600.}$$

The slack values are $x_3 = 200$ and $x_4 = x_5 = 0$. The non-basic variables are x_4 and x_5 , whose reduced costs -1 and -3 are strictly negative; therefore the optimal solution is unique.

Example 6: Solve by the simplex method:

$$\begin{aligned} &\text{maximize} && Z = x_1 - x_2 + 3x_3, \\ &\text{subject to} && x_1 + x_2 + x_3 \leq 10, \\ & && 2x_1 - x_3 \leq 2, \\ & && 2x_1 - 2x_2 + 3x_3 \leq 0, \\ & && x_1, x_2, x_3 \geq 0. \end{aligned}$$

Step 1: Convert the constraints into equations. All right-hand sides are non-negative and every constraint has the sign “ $\leq$ ”. Introduce slack variables $x_4, x_5, x_6 \geq 0$:

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 &= 10, \\ 2x_1 - x_3 + x_5 &= 2, \\ 2x_1 - 2x_2 + 3x_3 + x_6 &= 0. \end{aligned}$$

The standardized objective function is

$$Z = x_1 - x_2 + 3x_3 + 0x_4 + 0x_5 + 0x_6.$$

Step 2: Find the initial B.F.S. and identify degeneracy. The slack-variable columns form I_3 . Setting $x_1 = x_2 = x_3 = 0$ gives

$$x_4 = 10, \quad x_5 = 2, \quad x_6 = 0.$$

Thus the initial basis is $\{x_4, x_5, x_6\}$. It is feasible, but it is *degenerate* because the basic variable x_6 is zero.

The objective coefficients are

$$(c_1, c_2, c_3, c_4, c_5, c_6) = (1, -1, 3, 0, 0, 0).$$

Therefore the basic-cost vectors in the three tableau blocks are

Tableau block	Ordered basis	C_B
1	(x_4, x_5, x_6)	$(0, 0, 0)$
2	(x_4, x_5, x_3)	$(0, 0, 3)$
3	(x_2, x_5, x_3)	$(-1, 0, 3)$

This illustrates that c_{B_i} is copied from the objective coefficient of the i th current basic variable; it is not automatically zero after the initial tableau.

Step 3: Perform the simplex iterations. All iterations are shown in one combined table. Boxed entries are the pivot elements. We use $\Delta_j = c_j - c_B^T Y_j$, so a maximization problem is optimal when every $\Delta_j \leq 0$.

Table 14: Single combined simplex table for Example 8.10.3

B	c_B	x_B	1 x_1	-1 x_2	3 x_3	0 x_4	0 x_5	0 x_6	Minimum ratio Entering column
x_4	0	10	1	1	1	1	0	0	10/1 = 10
x_5	0	2	2	0	-1	0	1	0	Not eligible
x_6	0	0	2	-2	3	0	0	1	0/3 = 0 (minimum)
$Z = 0$		Δ_j	1	-1	3	0	0	0	x_3 enters
x_4	0	10	1/3	5/3	0	1	0	-1/3	10/(5/3) = 6 (minimum)
x_5	0	2	8/3	-2/3	0	0	1	1/3	Not eligible
x_3	3	0	2/3	-2/3	1	0	0	1/3	Not eligible
$Z = 0$		Δ_j	-1	1	0	0	0	-1	x_2 enters
x_2	-1	6	1/5	1	0	3/5	0	-1/5	—
x_5	0	6	14/5	0	0	2/5	1	1/5	—
x_3	3	4	4/5	0	1	2/5	0	1/5	—
$Z = 6$		Δ_j	-6/5	0	0	-3/5	0	-4/5	Optimal

First pivot: a degenerate pivot. Initially, $\Delta_3 = 3$ is the largest positive reduced cost, so x_3 enters. Only positive entries in the x_3 column are eligible:

$$\frac{10}{1} = 10, \quad \frac{0}{3} = 0.$$

The minimum ratio is zero, so x_6 leaves and the pivot is 3. Perform

$$R_3 \leftarrow \frac{1}{3}R_3, \quad R_1 \leftarrow R_1 - R_3, \quad R_2 \leftarrow R_2 + R_3,$$

where the last two operations use the normalized third row. Although x_3 enters the basis, its new value is zero. Hence the basis changes but the objective value remains $Z = 0$. This is called a *degenerate pivot*.

Second pivot. In the second block, $\Delta_2 = 1 > 0$, so x_2 enters. The only positive entry in its column is $5/3$ in the x_4 row. Thus

$$\frac{10}{5/3} = 6,$$

so x_4 leaves and the pivot is $5/3$. Perform

$$R_1 \leftarrow \frac{3}{5}R_1, \quad R_2 \leftarrow R_2 + \frac{2}{3}R_1, \quad R_3 \leftarrow R_3 + \frac{2}{3}R_1,$$

where the last two operations use the normalized first row.

In the final block, every reduced cost satisfies $\Delta_j \leq 0$; therefore the solution is optimal. Reading the basic variables gives

$$\boxed{x_1 = 0, \quad x_2 = 6, \quad x_3 = 4, \quad Z_{\max} = 6.}$$

The slack variables are $x_4 = 0$, $x_5 = 6$ and $x_6 = 0$. The non-basic variables x_1, x_4, x_6 have strictly negative reduced costs, so the optimal solution is unique.